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Tugas Jobsheet 9

Program:

Surat20.java:

```
package Jobsheet9;
```

```
public class Surat20 {
```

```
    String idSurat;
```

```
    String namaMahasiswa;
```

```
    String kelas;
```

```
    char jenisIzin;
```

```
    int durasi;
```

```
    Surat20(){
```

```
    }
```

```
    Surat20(String idSurat, String namaMahasiswa, String kelas, char jenisIzin, int durasi){
```

```
        this.idSurat = idSurat;
```

```
        this.namaMahasiswa = namaMahasiswa;
```

```
        this.kelas = kelas;
```

```
        this.kelas = kelas;
```

```
        this.jenisIzin = jenisIzin;
```

```
        this.durasi = durasi;
```

```
    }
```

```
}
```

StackSurat20.java:

```
package Jobsheet9;
```

```
public class StackSurat20 {
```

```
    Surat20[] stack;
```

```
    int size;
```

```
    int top;
```

```
    StackSurat20(int size){
```

```
        this.size = size;
```

```
        stack = new Surat20[size];
```

```
        top = -1;
```

```
    }
```

```
    boolean isfull(){
```

```
        if (top == size - 1) {
```

```
            return true;
```

```
        }else{
```

```
            return false;
```

```
        }
```

```
    }
```

```
    boolean isEmpty(){
```

```
        if (top == -1) {
```

```
            return true;
```

```
        }else{
```

```
            return false;
```

```
        }
```

```
    }
```

```

void push(Surat20 surat){
    if (!isfull()) {
        top++;
        stack[top] = surat;
    }else{
        System.out.println("Data Sudah Penuh!");
    }
}

```

```

Surat20 pop(){
    if (!isEmpty()) {
        Surat20 s = stack[top];
        top--;
        return s;
    }else{
        System.out.println("Tidak Terdapat Data");
        return null;
    }
}

```

```

boolean cariSurat(String namaSiswa){
    boolean isFound = false;
    for (int i = 0; i <= top; i++) {
        if (stack[i].namaMahasiswa.equalsIgnoreCase(namaSiswa)) {
            isFound = true;
        }
    }
    return isFound;
}

```

```

Surat20 LihatTerakhir(){
    if (!isEmpty()) {
        return stack[top];
    }else{
        System.out.println("Tidak Terdapat Data");
        return null;
    }
}
}
}

```

SuratDemo20.java:

```

package Jobsheet9;

import java.util.Scanner;

public class SuratDemo20 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        StackSurat20 stack = new StackSurat20(3);
        int pilih;

        do {
            System.out.println("=== Menu ===");
            System.out.println("1. Terima Surat Izin");
            System.out.println("2. Proses Surat Izin");
            System.out.println("3. Lihat Surat Izin Terakhir");
            System.out.println("4. Cari Surat");
            System.out.print("Pilih Menu:");
            pilih = sc.nextInt();

```

```
sc.nextLine();
```

```
switch (pilih) {
```

```
    case 1:
```

```
        System.out.print("ID Surat: ");
```

```
        String ID = sc.nextLine();
```

```
        System.out.print("Nama: ");
```

```
        String nama = sc.nextLine();
```

```
        System.out.print("Kelas: ");
```

```
        String kelas = sc.nextLine();
```

```
        System.out.print("Jenis Izin(S/I): ");
```

```
        char izin = sc.findInLine(".").charAt(0);
```

```
        System.out.print("Durasi: ");
```

```
        int durasi = sc.nextInt();
```

```
        sc.nextLine();
```

```
        Surat20 surat = new Surat20(ID, nama, kelas, izin, durasi);
```

```
        stack.push(surat);
```

```
        break;
```

```
    case 2:
```

```
        Surat20 proses = stack.pop();
```

```
        if (proses != null) {
```

```
            System.out.println("Memproses Surat Dari " + proses.namaMahasiswa);
```

```
            System.out.print("Input Status (Terima / Tidak Diterima): ");
```

```
            String status = sc.nextLine();
```

```
            System.out.printf("Surat %s %s\n", proses.namaMahasiswa, status);
```

```
        }
```

```
        break;
```

```
    case 3:
```

```
        Surat20 lihat = stack.LihatTerakhir();
```

```

        if (lihat != null) {
            System.out.println("Surat terakhir berasal dari " + lihat.namaMahasiswa);
        }
        break;
    case 4:
        System.out.print("Input nama Mahasiswa: ");
        String nama2 = sc.nextLine();
        if (stack.cariSurat(nama2)) {
            System.out.println("Data Ditemukan");
        } else {
            System.out.println("Data Tidak Ditemukan");
        }
        break;
    default:
        break;
    }
} while (pilih >= 1 && pilih <= 4);
sc.close();
}
}

```

Hasil:

```
=== Menu ===
1. Terima Surat Izin
2. Proses Surat Izin
3. Lihat Surat Izin Terakhir
4. Cari Surat
Pilih Menu:1
ID Surat: 1
Nama: akmal
Kelas: 1g
Jenis Izin(S/I): s
Durasi: 2
=== Menu ===
1. Terima Surat Izin
2. Proses Surat Izin
3. Lihat Surat Izin Terakhir
4. Cari Surat
Pilih Menu:1
ID Surat: 2
Nama: fauzi
Kelas: 1g
Jenis Izin(S/I): i
Durasi: 4
=== Menu ===
1. Terima Surat Izin
2. Proses Surat Izin
3. Lihat Surat Izin Terakhir
4. Cari Surat
Pilih Menu:1
ID Surat: 3
Nama: reyhan
Kelas: 1g
Jenis Izin(S/I): i
Durasi: 6
```

```
=== Menu ===
1. Terima Surat Izin
2. Proses Surat Izin
3. Lihat Surat Izin Terakhir
4. Cari Surat
Pilih Menu:4
Input nama Mahasiswa: reyhan
Data Ditemukan
=== Menu ===
1. Terima Surat Izin
2. Proses Surat Izin
3. Lihat Surat Izin Terakhir
4. Cari Surat
Pilih Menu:3
Surat terakhir berasal dari reyhan
=== Menu ===
1. Terima Surat Izin
2. Proses Surat Izin
3. Lihat Surat Izin Terakhir
4. Cari Surat
Pilih Menu:2
Memproses Surat Dari reyhan
Input Status (Terima / Tidak Diterima): Tidak Diterima
Surat reyhan Tidak Diterima
=== Menu ===
1. Terima Surat Izin
2. Proses Surat Izin
3. Lihat Surat Izin Terakhir
4. Cari Surat
Pilih Menu:
```